

The Use of Artificial Intelligence to Identify Predatory Litigation in Brazil

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Abstract

This article investigates the application of artificial intelligence (AI) to address repetitive and abusive litigation in Brazilian courts. In recent years, the Brazilian judiciary has experienced a surge in predatory litigation, leading to increased case backlogs and inefficiencies. Leveraging AI tools for litigation identification and classification presents a promising solution to mitigate these challenges. This study comprehensively reviews existing research and AI-driven tools deployed within the Brazilian judicial system to combat repetitive claims, mainly focusing on initiatives like the Bastião Project, Genesis, and Sherlock systems. By examining AI's technological approaches, challenges, and efficacy in managing predatory litigation, this paper highlights AI's potential to enhance judicial efficiency and integrity. The findings underscore the necessity of continuous refinement in AI models to address ethical considerations and data transparency and adapt to evolving judicial demands.

Keywords

Artificial Intelligence in Legal Analysis, Machine Learning Efficiency, Document Classification Automation, Data-Driven Legal Processes, Repetitive Themes

Introduction

The increasing complexity of legal disputes and the growing number of court cases exert significant pressure on judicial systems worldwide. In Brazil, one notable challenge is the proliferation of repetitive and abusive litigation, which substantially burdens the judiciary, affecting both efficiency and accessibility. Repetitive claims, often driven by predatory litigants, can consume judicial resources unnecessarily, delaying the resolution of other cases and impacting fair access to justice. Brazil's National Council of Justice (CNJ) has identified this issue as a critical concern, advocating for solutions that improve judicial efficiency while preserving the right to due process.

According to the CNJ report “Justice in Numbers 2024” (1), by the end of 2023, Brazilian courts had 83.8 million pending cases, representing a 1.1% increase compared to the end of 2022. When excluding the 18.5 million suspended cases, the number of ongoing proceedings amounts to 63.6 million. The influx of new cases reached a record high in 2023, with 35.3 million cases filed, marking a 9.4% increase from the previous year. Of these, 22.6 million were first-time filings, excluding appeals and enforcement proceedings. Brazilian courts rendered judgments in 33.2 million cases in 2023, the highest figure recorded in the historical series. This total represents an 11.3% increase compared to 2022 and a cumulative growth of 40.3% over the past 14 years. As observed, the CNJ's statistical reports have consistently highlighted a persistent rise in the accumulation of pending cases, despite successive productivity records. The São Paulo Court of Justice (TJSP) stands as the largest court in Brazil in terms of size and case volume. In 2023 alone, 6,863,658 new lawsuits were filed, while 24,287,179 cases remained

pending trial. The court's workforce comprises 2,656 judges and 56,185 staff members (1).

An analysis of the civil litigation landscape reveals that a significant portion of disputes is concentrated among a recurring group of habitual litigants (2), particularly government entities and financial institutions, which together account for more than 50% of filed lawsuits (3). In general, the scenario indicates a growing volume of litigation, largely driven by the same group of actors - notably government bodies and suppliers of goods or services — whose mass relationships have a high potential to generate widespread conflicts. These conflicts often result in the filing of numerous similar lawsuits, giving rise to repetitive or mass litigation. Since 2020, the judiciary has experienced a continuous increase in pending cases, with 896,000 additional cases recorded between 2022 and 2023, representing a 1.1% growth (1). The accumulation of unresolved cases continues to rise year after year. While excessive litigation is a multifaceted issue with various underlying causes, one contributing factor is the rise of repetitive and abusive litigation.

In recent years, artificial intelligence (AI) advancements have offered promising tools for addressing these challenges in the legal sector. AI-driven solutions, particularly those employing machine learning and natural language processing, enable the automatic classification, filtering, and prioritisation of legal cases. Such technologies can

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support judges and legal professionals by identifying patterns of abusive litigation, flagging high-risk cases early, and facilitating data-driven decision-making processes. Several Brazilian courts have begun implementing AI tools to streamline judicial workflows and manage high-volume, repetitive demands. Notable projects like the Bastião Project and the Sherlock System utilise AI to detect potentially harmful litigation practices, promoting justice efficiency and reducing system congestion.

In the context of adopting new technologies in the legal field, a study conducted by the Center for Innovation, Administration, and Research of the Judiciary at the Getúlio Vargas Foundation (FGV), under the coordination of Justice Luis Felipe Salomão, surveyed national initiatives and experiences involving the application of AI in Brazilian courts in 2020 (4). The study identified several key objectives achieved through the implementation of AI, including the optimization of services for lawyers and the public, increased security, automation of routine activities, improved human resource management for core judicial functions, and greater efficiency in procedural workflows (4).

In 2023, the CNJ conducted a follow-up survey to provide an updated overview of AI usage within the Brazilian judiciary. The survey revealed that neural networks are the most commonly used algorithms, appearing in 79 projects (56.4%), reflecting their versatility and capacity for natural language processing — both essential for managing the judiciary's extensive data volume (5). The survey also highlighted three main challenges faced in implementing AI projects in courts: the scarcity of qualified professionals to work with AI; difficulties in accessing adequate data (in terms of quantity, quality, and diversity); and the complexity of integrating AI with existing systems (5). AI projects within the judiciary encompass a wide range of applications, with notable emphasis on case similarity detection (mentioned 69 times, in 49.3% of projects), document classification (mentioned 66 times, in 47.1% of projects), and the automation of procedural documents (mentioned 39 times, in 27.9% of projects) (5). Additionally, the identification of predatory litigation was cited in 15 instances, representing 10.7% of the projects, as detailed in the 2023 Artificial Intelligence Research Panel (6).

This study aims to investigate existing research and technological tools that leverage AI to combat repetitive and abusive litigation in Brazilian courts. By comprehensively reviewing scientific literature and technical documentation, we explore the applications, limitations, and future directions for AI-driven solutions in mitigating litigation abuse within Brazil's legal framework. Our findings contribute to a deeper understanding of AI's role in enhancing judicial processes and highlight key areas where technological interventions can reinforce the integrity and efficiency of the judiciary.

The structure of this article is as follows: Section provides the theoretical foundation, defining predatory litigation and situating it within the broader context of legal and economic impacts. Section reviews the scientific literature, discussing current methodologies for identifying and combating abusive litigation. Section details recent AI initiatives within Brazilian courts aimed at improving judicial efficiency. Finally, Section synthesizes the findings,

and Section concludes with a summary and future research recommendations.

Theoretical Reference

The concept of *predatory litigation*, also referred to as *sham litigation*, has received considerable attention within legal and economic disciplines due to its adverse implications for judicial efficiency and market competition. According to Salgado et al.(7), predatory litigation is characterised by the strategic misuse of legal processes to gain an unfair competitive advantage, typically by burdening competitors with excessive legal costs or delaying their market entry. This litigation category is particularly prevalent in sectors dominated by intellectual property rights, such as technology and pharmaceuticals, where litigation is often weaponised to exclude rivals without substantive legal grounds (8; 9).

In Brazil, predatory litigation presents a significant and growing challenge, contributing to inefficiencies and costs within the judiciary. Reports by the CNJ indicate that a high volume of predatory cases imposes undue strain on judicial resources, undermining goals such as National Goal 1, which seeks to resolve more cases than are filed annually (10). The CNJ highlights recurring indicators of predatory litigation, including repetitive lawsuits with tenuous legal bases and the deliberate use of judicial proceedings to delay obligations or extract unfair settlements (11; 12).

From a theoretical perspective, predatory litigation can be understood as a form of *non-price competition*, where firms exploit judicial processes rather than competing based on product quality or pricing strategies (13). Christopher C. Klein(9) provide a foundational economic analysis, describing predatory litigation as a strategy in which the indirect benefits of engaging in litigation—such as deterring competitors or exhausting their financial resources—outweigh the expected outcomes of winning the case. This strategic misuse of litigation, often pursued with minimal expectations of success, has been widely observed across studies on anticompetitive behaviours (14).

Felipe Albertini Nani Viano(15) further advance the understanding excessive litigation through a structured framework that categorises litigation behaviours. Among the various forms of excessive litigation, two phenomena stand out due to their anomalous nature: *repetitive litigation* and *predatory litigation*.

Repetitive litigation

Repetitive litigation, unlike predatory litigation, is not inherently abusive. On the contrary, it is a natural consequence of mass societies, arising from widespread violations of rights that affect large groups of people. It is characterized by the repeated filing of similar claims, often based on the same legal theory or stemming from a common source of conflict, aimed at enforcing legitimate rights. The recurrence of lawsuits involving the same subject matter is a distinctive feature of the Brazilian legal system, whose procedural framework — predominantly private and individualistic in nature — was designed to resolve isolated disputes between parties without consideration for collective treatment of similar issues (16). Certain characteristics of litigation in Brazil, such as recurring violations of rights by

habitual litigants that impact large populations, the relatively low cost of accessing the judiciary, and the prominent role of the courts, have contributed to a significant volume of repetitive claims. While the emergence of repetitive lawsuits is an expected consequence of these factors, some judges began to observe exaggerated patterns in certain cases. They identified numerous lawsuits with identical or highly similar complaints, the same factual narratives, filed by the same lawyers or legal groups, and following specific patterns — including requests for free legal aid, exemption from conciliation hearings, and claims involving plaintiffs residing in distant regions. One of the earliest judicial decisions to address this phenomenon was issued in late 2011 (17):

Over the past year, more than a thousand lawsuits were filed in the Santo Amaro District Court, all presenting identical causes of action and claims. The legal arguments included: (a) the nonexistence of a contract between the plaintiff and the defendant, and (b) unlawful inclusion of the plaintiff's name in a debtor registry. The relief sought was: (a) a declaration of the non-existence of the debt, and (b) compensation for moral damages. (...) The unusual similarities drew attention to several aspects: (a) the coordinated actions of four lawyers, who were confirmed to be from the same law firm; (b) the fact that the complaints were entirely disconnected from the actual facts, amounting to contempt of court; and (c) suspicions of ethical misconduct by the lawyers.

Upon further investigation, judges began to notice recurring patterns across many of these lawsuits. In some instances, the factual narrative presented in the complaint was completely at odds with the reality of the case. In others, it was discovered that plaintiffs were unaware of or had no interest in the lawsuits filed on their behalf. More concerning were cases involving forged documents submitted in court.

These practices were not isolated incidents; instead, they revealed a deliberate *modus operandi* aimed at exploiting the judicial system through abusive filings. This phenomenon became known as predatory litigation (18), or aggressive litigation (19). Predatory litigation is a specific form of abusive litigation. According to Recommendation No. 159/2024 (20) issued by CNJ, abusive litigation is defined as conduct that deviates from or manifestly exceeds the social, legal, political, or economic purposes of the right to judicial access, including actions that compromise the judiciary's ability to deliver justice.

Predatory litigation

Predatory litigation, in particular, is characterized by the mass filing of lawsuits or procedural actions based on standardized complaints that often omit or distort facts. These lawsuits are frequently filed without the knowledge or consent of the individuals named as plaintiffs and, in some cases, involve the use of falsified documents. This practice represents a distortion of the concept of access to justice, often facilitated by the misuse of free legal aid mechanisms, allowing for the initiation of lawsuits without genuine disputes — a phenomenon that can be described as fake

litigation (21). In addition to harming the opposing party, predatory litigation drains judicial resources, contributing to delays and court congestion. Fundamentally, it is a strategy for the abusive and repetitive use of the judiciary. However, the mere volume of lawsuits is not sufficient to characterize a practice as predatory. If that were the case, all repetitive claims would be considered abusive, which is not the case. Predatory conduct involves the intentional misuse of judicial mechanisms, often with fraudulent or unethical tactics aimed at exploiting procedural advantages.

Predatory litigation is inherently systemic, multidimensional, and disruptive. The lawsuits arising from these disputes exploit judicial mechanisms for purposes extraneous to legitimate legal relief, manifesting as a deliberate strategy to delay proceedings, overburden opponents, and extract undue economic benefits. The CNJ has underscored the importance of transitioning towards *responsible litigation practices* through typological tools and diagnostic frameworks to identify and mitigate such behaviours.

Abusive litigation

Specifically, abusive litigation encompasses various subcategories, including:

- **Fraudulent Claims:** Fraudulent claims involve the deliberate submission of falsified or manipulated information to obtain undue legal advantages. Documented cases include lawsuits seeking the confirmation of a non-existent debt and claims for moral damages based on falsified documents, intentionally omitting prior annotations to circumvent Precedent No. 385 of the Brazilian Superior Court of Justice (STJ) (22). Other examples include cases where powers of attorney were altered, resulting in lawsuits being filed without the plaintiff's knowledge or consent (23).
- **Improbis litigation:** In international legal literature, improbis litigation is comparable to the concept of frivolous litigation, referring to claims filed without the necessary diligence or legal basis. Examples include lawsuits seeking confirmation of the absence of a debt based on the plaintiff's assertion that they "do not recall" the debt, despite being fully aware of its existence (24). Additionally, revisionary lawsuits are often filed against established precedents, without distinction, to exploit the opposing party's disorganization and assert rights that do not actually exist (25). Such actions may also seek to have legitimate legal relationships declared null or nonexistent.
- **Frivolous Demands:** Frivolous demands refer to claims devoid of substantive merit, often filed to burden both the judiciary and opposing parties. While sometimes used synonymously with improbis litigation, frivolous claims are distinguished by their deliberate fragmentation of low-value issues into separate lawsuits, significantly reducing the economic or social value of each claim. This strategy aims to generate or multiply financial gains for plaintiffs or their attorneys. For instance, preparatory actions,

such as requests for document disclosure, are frequently filed without prior administrative requests. Additionally, claims that could be consolidated into a single lawsuit are intentionally fragmented to take advantage of opposing parties' disorganization or to increase the attorney's fees awarded in each individual case (26).

- **Procrastinatory Lawsuits:** Procrastinatory lawsuits are primarily filed to delay inevitable legal or financial consequences. Although often associated with defendants' behavior, they can also arise from plaintiffs who misuse their right to legal action to postpone compliance with obligations. Common examples include lawsuits challenging debtor registry listings without any factual basis (27) or actions seeking to revise financial contracts with the sole purpose of halting the enforcement of guarantees without paying the undisputed portion of the debt (28). Other procrastinatory tactics involve filing delaying appeals or requesting procedural steps without paying the corresponding fees to avoid case suspension and prolong the litigation process, thereby increasing legal fees for ongoing case monitoring (29).
- **Procedural Harassment:** Procedural harassment refers to the abusive use of procedural tools to overwhelm opposing parties, escalate litigation costs, and disrupt judicial processes. An example is found in CNJ Recommendation No. 127, which addresses the misuse of litigation to suppress freedom of expression rights. Similarly, in Special Appeal No. 1.817.845, Justice Nancy Andrighi used the term to describe the filing of numerous lawsuits aimed at preventing the enforcement of a right already recognized in another legal proceeding (30).
- **Sham Litigation:** Sham litigation refers to the use of legal proceedings as a pretext to achieve extralegal objectives, such as damaging a competitor's market reputation (31). In a political context, sham litigation often manifests in the form of popular lawsuits aimed at tarnishing the public image of political adversaries through baseless accusations and legal harassment (32).
- **Spam Processual (Mass Filing of Baseless Cases):** Spam litigation involves the large-scale submission of template-based claims that impose disproportionate burdens on courts and defendants. This type of abuse is primarily intra-procedural, with mass filings of conditional requests or claims that blatantly contradict the procedural history of the cases (33). These template-based claims are often filed en masse across multiple cases, creating an administrative overload for the judiciary and delaying legitimate claims.

In addition to disproportionately consuming judicial resources, abusive litigation directly impacts society by undermining the judiciary's ability to provide timely access to justice for all citizens. One notable example is a study conducted in the Ribeirão Preto District, which revealed that a single group of lawyers filed 1,500 and 1,760 lawsuits in July and September 2016, respectively, averaging 28 new cases per day, with peaks of 100, 160, and even 200 lawsuits

in a single day (34). These cases accounted for 38% of the total civil case filings in the district. The financial impact was significant, with unpaid court fees due to free legal aid amounting to approximately R\$110,712,019.00. The influx of these cases contributed to a substantial increase in the average time required for case resolution, from 364 days in 2012 to 930 days in 2016 (35). Similar phenomena were identified in other states. In Mato Grosso do Sul, for example, 64,037 lawsuits related to consigned loans were filed, with evidence of mass abuse in 34,471 of them, all filed by a single group of lawyers (36). Upon analyzing a sample of 300 cases, it was found that 100% of the complaints were hypothetical, and 80% resulted in the dismissal of claims along with a finding of contempt of court. Reports of similar abusive practices have been documented in several other states through the Abusive Litigation Information Network established by the National Council of Justice (37).

NUMOPEDE

In response to these growing concerns, TJSP created the Center for Monitoring Demand Profiles (NUMOPEDE) in 2016 to track and manage lawsuits that, due to their characteristics, have a substantial impact on judicial services (38). The objective of NUMOPEDE is to monitor major case types, identify frequent litigants and their attorneys by district, and develop strategies to mitigate the effects of mass or repetitive litigation, thereby preventing abusive use of the judiciary.

In its first year, NUMOPEDE-TJSP published a significant study in its Bimonthly Activity Report (October–November 2016), which focused on atypical judicial practices involving lawsuits related to document disclosure, unlawful debtor registry entries, payment consignments, and the duty to provide information (39).

While initial studies centered on consumer lawsuits, subsequent research revealed other forms of predatory intra-procedural conduct. These included mass filings of petitions unrelated to the subject matter, as well as abusive practices by suppliers against consumers and even between companies. In some instances, consumers were used as intermediaries to create artificial demands or impose unilateral contract terms, such as specifying unnecessary treatments or materials from particular brands in health contracts (40).

NUMOPEDE's Communiqué No. 02/2017 outlined key indicators of abusive use of the judiciary and has since served as a reference for identifying and addressing predatory litigation (41). Over the years, NUMOPEDE has refined its workflow to better detect and manage abusive litigation, establishing best practices to address such issues (42).

NUMOPEDE's identification process involves administrative information management and is based on data provided through a dedicated email system, which is manually compiled and categorized by primary themes. The center also analyzes metadata from the digital case management system, examining information such as case type, subject matter, monetary value, key litigants, and their respective attorneys by district.

This manual analysis of filing patterns and metadata aims to detect atypical behaviours, such as spikes in case filings (high volumes of lawsuits filed in a short

period), fragmentation of claims (multiple related claims that could be consolidated into a single lawsuit), and case dispersion (filing cases across multiple jurisdictions to obscure connections between claims).

Once atypical patterns are identified, NUMOPEDE conducts a sample analysis of case files to detect repeated behaviors and characteristics. The findings are then used to develop best practices, which are shared with judges, the Public Prosecutor's Office, and the Brazilian Bar Association when evidence of criminal or disciplinary misconduct is uncovered. This model has been replicated by several other courts in Brazil (43).

Over the years, NUMOPEDE has gathered a substantial body of data on predatory litigation practices, many of which have been confirmed by final court rulings. These cases often lead to disciplinary proceedings against the attorneys involved, and in more severe instances, result in criminal charges. Based on this data, NUMOPEDE has produced two major studies on predatory litigation: one focusing on procedural activities and the other on estimating the financial impact on the public treasury (42).

The first study was based on two criteria: (i) procedural activity arising from cases flagged through reports submitted to NUMOPEDE, and (ii) procedural activity measured according to the methodology outlined in Technical Note No. 01/2022 from the Minas Gerais Court of Justice's Center for Innovation in Judicial Management (CITJMG). The first criterion for identifying predatory litigation considered communications received by NUMOPEDE between 2016 and 2021. It focused on the number of lawsuits filed by specific procedural actors (based on metadata such as plaintiff/attorney) referenced in these communications, each classified as a "case" and analyzed multiple times over the years. Based on this criterion, an analysis of the 30 cases with the highest number of studies conducted by NUMOPEDE during the period revealed a procedural volume of approximately 120,000 lawsuits over six years, impacting more than 840 judicial bodies.

Applying the same average procedural volume to 503 additional cases identified through communications, it was projected that predatory litigation generated approximately 337,000 lawsuits annually during the period analyzed, affecting nearly all civil courts. The second criterion, based on monitoring conducted by the Intelligence Center of the Court of Justice of Minas Gerais (CITJMG), suggested an even higher volume of predatory litigation, accounting for up to 30% of all procedural activity. Analyzing the year-on-year variation in new lawsuits filed between 2016 and 2021 (excluding tax executions, criminal cases, and cases in the Special Civil Courts), it was found that predatory litigation generated an average of 605,000 lawsuits per year during this period. In the year with the highest number of civil case filings, more than 722,000 new predatory lawsuits were identified. Even in the year with the lowest number of filings, the volume exceeded 448,000 lawsuits with predatory characteristics.

Beyond the impact on procedural activity, NUMOPEDE conducted a study to estimate the financial damage caused to the public treasury. It is important to note that calculating the cost of a lawsuit to the public treasury is one of the greatest

challenges in judicial research, with various methodologies available for this purpose.

Two criteria were used to estimate these costs:

1. The cost per lawsuit, as estimated in a study conducted by the Institute for Applied Economic Research (IPEA).
2. The cost per lawsuit, as calculated under CNJ Resolution No. 76/2009, based on the Total Justice Expenditure (DPJ) indicator divided by the number of new cases filed in the Sao Paulo Court of Justice (TJSP), excluding tax executions, criminal actions, and Special Civil Court cases.

Based on the first criterion — the cost per lawsuit estimated by the IPEA study at R\$8,270.00 (adjusted to March 2022) — the estimated financial loss caused by 30 procedural actors reached approximately R\$999.3 million over the period analyzed, or about R\$166.5 million annually. Expanding this estimate to the procedural volume of 503 cases analyzed by NUMOPEDE between 2016 and 2021, the financial loss to the public treasury was projected at approximately R\$16.7 billion over the period, or R\$2.7 billion per year.

Further, using the CITJMG criterion — which estimates that 30% of procedural activity in civil courts is attributable to predatory litigation — the financial loss to the public treasury was estimated at R\$4.4 billion for the period analyzed. When calculating the cost per new lawsuit based on CNJ Resolution No. 76/2009, the financial impact of predatory litigation in civil courts was estimated at R\$1.9 billion in 2017 and R\$1.2 billion in 2021, assuming 30% of new lawsuits were related to predatory practices.

Judiciary Intelligence Centers

In response to the growing prevalence of repetitive and mass litigation, CNJ Resolution No. 349/2020 mandated the creation of Judiciary Intelligence Centers (CIPJs) to "identify and propose appropriate strategic measures to address repetitive and mass litigation in the Brazilian judiciary" (44). Under this resolution, the CIPJs are tasked with preventing the filing of repetitive lawsuits by identifying their root causes, promoting self-resolution mechanisms, or referring matters to administrative bodies for appropriate solutions. The centers are also responsible for proposing technical notes to improve legislation. CIPJs operate primarily as administrative, non-judicial bodies, focusing on three main areas:

1. Demand Prevention – Identifying and addressing the causes of repetitive lawsuits before they are filed.
2. Procedural Monitoring – Collecting and analyzing procedural information to identify patterns of abusive litigation.
3. Precedent Management – Ensuring consistent application of legal precedents to discourage repetitive filings. With the creation of CIPJs, the issue of predatory litigation began to be addressed at both national and local levels (45).

Despite the establishment of initiatives like NUMOPEDE and CIPJs, the analysis of judicial data to detect abusive

litigation practices still relies heavily on manual processes. This dependence on manual analysis limits the judiciary's ability to proactively and efficiently address predatory claims.

In this context, information technology solutions, particularly artificial intelligence, have emerged as valuable tools to revolutionize the identification and management of predatory litigation. AI's ability to identify patterns, automate tasks, and optimize resource allocation offers significant potential to enhance judicial efficiency.

The application of AI in this field is not intended to replace human analysis but rather to complement and enhance it. By automating routine tasks and improving data analysis capabilities, AI allows judicial bodies to allocate resources more effectively and promote a faster, more efficient judicial process.

Concrete measures have been proposed in the Brazilian context, where systemic judicial congestion exacerbates the challenges posed by predatory litigation. We can cite enhanced procedural scrutiny for mass filings, the prevention of unjustified requests for interim relief, and the promotion of alternative dispute resolution mechanisms to alleviate reliance on formal litigation processes. Internationally, similar concerns have been raised. The European Commission, for example, has recognised the misuse of judicial proceedings as a tool for distorting competition. In the United States, such practices are particularly prominent within the technology sector, where patent litigation is strategically employed as a barrier to entry (46).

CNJ regulations and recommendations

The phenomenon of predatory litigation has gained significant attention from Brazil's CNJ in recent years, driven by the need to mitigate its negative impact on the efficiency of the judicial system. In response to this growing concern, the National Internal Affairs Office of the CNJ approved Strategic Guideline No. 7 in 2023 (47), followed by Strategic Guideline No. 6 in 2024 (48). These guidelines direct Brazilian courts to focus their efforts on the following objectives:

1. Regulating and promoting practices and protocols to combat predatory litigation, with a preference for developing information technology solutions to monitor ongoing cases.
2. Transferring relevant data to the National Internal Affairs Office to support the creation of a unified monitoring panel specifically designed to track and address abusive litigation practices.

Building on the advancements introduced by these guidelines, the CNJ established the Abusive Litigation Information Network, coordinated by the National Internal Affairs Office. This network functions as a collaborative platform for courts to share experiences, exchange knowledge, and disseminate best practices. It reinforces the importance of integrated and standardized approaches across the judiciary to effectively address the issue of predatory litigation (37).

The CNJ Recommendation No. 159 of 23 October 2024 (20) provides a comprehensive and actionable

framework to address predatory litigation within Brazil. This directive outlines specific behaviours indicative of abusive practices (Annex A(20)), judicial actions to mitigate such cases (Annex B(20)), and structural recommendations for courts to enhance detection and management of predatory behaviours (Annex C(20)). The adoption of AI-driven tools to monitor litigation patterns and identify mass filings further reflects the judiciary's commitment to optimising resources and ensuring procedural efficiency (49).

In summary, predatory litigation goes beyond being just a legal irregularity; it is a systemic issue that weakens judicial efficiency and disrupts economic fairness. Tackling this challenge demands a comprehensive approach that brings together new legislation, proactive judicial oversight, and innovative technologies. The use of artificial intelligence in judicial processes offers a promising solution, helping to identify abusive patterns, speed up case resolution, and create a fairer and more efficient legal system for all.

Literature Review

This section outlines the research methodology used for mapping studies focused on identifying and combating predatory litigation in Brazil. Following the established guidelines of Barbara Kitchenham and Charters Stuart (50), Petersen et al. (51), and brings2018 et al. (52), we adopted a structured framework to ensure thorough and reliable findings. The following sections detail the study's design, including our objectives, research questions, and the search protocol developed to capture relevant literature and case studies within the Brazilian legal context.

Research questions

This study is centred around the following research question (RQ):

RQ: *What are the current methods and tools used to identify predatory or abusive litigation lawsuits in Brazil?*

To address this research question, we have divided it into seven specific questions. The answers to these questions will enable us to classify and organize the documents for further analysis. The specific research questions (SRQs) are listed below.

SRQ1: *How has artificial intelligence been applied to combat or mitigate predatory litigation in legal systems, particularly Brazil?*

SRQ2: *What machine learning approaches have proven effective in identifying patterns of abusive litigation in lawsuits?*

SRQ3: *How do AI-based methods compare to traditional human analysis and manual case review in detecting predatory litigation?*

SRQ4: *What challenges and limitations are AI tools facing in addressing predatory litigation within the Brazilian legal context?*

SRQ5: *What future opportunities exist for improving AI-driven solutions in the fight against abusive litigation in Brazil's judiciary?*

Search protocol

This section outlines the search protocol initially developed by the first author and subsequently reviewed by the second and third authors. The scope of this study included the search terms defined by Kitchenham et al. (2007): Population, Intervention, Comparison, Outcomes, and Context (PICOC). These terms were used to identify relevant keywords and formulate search strings based on our research questions.

Population: judicial cases, predatory litigation, abusive litigation.

Intervention: artificial intelligence, machine learning, legal technology.

Comparison: traditional methods, human analysis, manual case review.

Outcomes: Methods, tools, and approaches to identify or combat predatory litigation.

Context: Brazilian courts of justice.

The keywords identified and their respective synonyms are shown in Table 1. We defined a search using the following base search string:

((“predatory litigation” OR “abusive litigation” OR “vexatious litigation” OR “litigância predatória” OR “litigância abusiva”) AND (“artificial intelligence” OR “AI” OR “inteligência artificial” OR “machine learning” OR “aprendizado de máquina” OR “legal technology” OR “tecnologia jurídica”) AND (“litigation detection” OR “identificação de litigância” OR “litigation identification” OR “detecção de litígios” OR “ferramentas para identificar litígios” OR “tools for identifying lawsuits” OR “combating abusive litigation” OR “combate à litigância abusiva”) AND (“Brazil” OR “Brasil”))

We needed to limit the number of keywords to make the document search more general. This study focused on documents published between 2014 and 2024, written in either Portuguese or English, within Computer Science or Law. The search process for this study utilized an automated search of indexing systems and digital libraries. Table 2 presents the search queries and corresponding databases used.

We started with a preliminary search string that generated a list of documents from each database considered. The Scopus database returned no results, Google Scholar provided 73 documents, and Consensus yielded nine documents. Additionally, we manually included three more documents. Next, we utilized an online tool called Parsifal (56) to import the collected documents’ references and eliminate duplicates. We retrieved 85 documents, with seven duplicates, leaving us with 82 unique documents. For the first round of filtering, the first author reviewed the titles, keywords, and abstracts of the documents and applied the inclusion and exclusion criteria outlined in Tables 3 and 4. We also considered the criteria related to publication year,

language, and subject area again, as some databases do not allow these filters during searches.

Documents that met at least one inclusion criterion were included in the study, while those that met any exclusion criterion were excluded. After applying the first filter, 35 documents remained, which were then passed on to the second filter. The quality assessment of these primary documents was conducted through a thorough reading by the first author. To evaluate quality, we developed six questions, each answered with one of the following values: “Yes”, “Partially”, or “No”. Each question/answer pair carries a specific score, as detailed in Table 5. Consequently, each document could earn between 0 and 14 points based on the six questions. We established a cut-off score of 8 points; therefore, 26 documents that scored above this threshold had their primary information recorded, as shown in Table 6.

Ultimately, the second and third authors reviewed each analysis activity to ensure consistency in the analysis and consolidation of the results. The final list of mapped documents is as follows:

• Enfrentamento da Litigância Predatória no Poder Judiciário do Rio Grande do Norte (57)

- **Authors:** Carlos Romero Bacurau de Brito
- **Year:** 2024
- **Methodology:** Manual Review
- **Design:** Systematic Review
- **Findings:** The study concludes that implementing measures from the Technical Note No. 01/2020 issued by the Intelligence Center of the Special Courts in Rio Grande do Norte has shown initial positive impacts in reducing regional predatory litigation. Statistical data reveal a decline in cases of undue inclusion in the debtor registry, suggesting that these measures have deterred abusive claims.
- **Limitations:** The study points out that while preliminary data indicate success, further adherence monitoring and analysis are needed to substantiate long-term effectiveness.
- **Suggestions for Future Research:** Future research should focus on a more detailed adherence analysis to better assess the long-term impact of the implemented measures and explore improvements in AI and digital tools to streamline predatory litigation identification and prevention.

• Acesso Inautêntico à Justiça e a Crise da Jurisdição: As Taxas Processuais na Litigância Predatória (58)

- **Authors:** José Laurindo de Souza Netto, Adriane Garcel, Karen Paiva Hippertt
- **Year:** 2022
- **Methodology:** Manual Review
- **Design:** Theoretical Analysis
- **Findings:** Brazil’s current procedural cost system encourages habitual and frivolous litigation, which strains judicial resources and hinders equitable access to justice. The authors advocate for reformed fee regulations to

Table 1. Keywords and synonyms

Keyword	Synonyms
Artificial Intelligence	AI, AI-driven solutions, legal AI tools, machine learning, supervised learning
Combatting Abusive Litigation	addressing frivolous claims, mitigating predatory lawsuits, reducing vexatious litigation
Judicial Cases	court cases, lawsuit, legal cases, litigation
Legal Technology	AI-driven legal tools, law tech, legal tech
Litigation Detection	lawsuit detection, legal pattern recognition, litigation identification
Predatory litigation	abusive litigation, frivolous lawsuits, vexatious litigation
Traditional Legal Methods	conventional legal practices, human case examination, human legal analysis, manual case review, manual litigation assessment

Table 2. Database and search sentences of the initial collect of documents.

Database	Search string
Scopus (53)	(TITLE-ABS-KEY (“litigância predatória” OR “litigância abusiva” OR “predatory litigation” OR “abusive litigation”) AND TITLE-ABS-KEY (“artificial intelligence” OR “machine learning” OR “legal technology” OR “inteligência artificial” OR “aprendizado de máquina” OR “tecnologia jurídica”))
Google Scholar (54)	(“litigância predatória” OR “litigância abusiva” OR “predatory litigation” OR “abusive litigation”) AND (“artificial intelligence” OR “machine learning” OR “legal technology” OR “inteligência artificial” OR “aprendizado de máquina” OR “tecnologia jurídica”)
Consensus (55)	Search scientific document databases and Brazilian court websites for real cases of the use of Artificial Intelligence to identify or combat legal actions considered predatory or abusive litigation.

Table 3. Inclusion criteria for the selection documents.

Inclusion criterion
Documents published in the last 10 years.
Documents presenting real-world applications of AI or legal technology in the detection or prevention of predatory litigation.
Documents written in English or Portuguese.
Research discussing using Artificial Intelligence, Machine Learning, or Legal Technology in identifying or combatting predatory litigation, abusive litigation, or frivolous lawsuits.
Documents focused on Brazil or involving Brazilian legal systems.

Table 4. Exclusion criteria for the selection documents.

Exclusion criterion
Documents have not been published in the last 10 years.
Documents not available in English or Portuguese.
Not accessible.
Documents that do not focus on Brazil or do not include Brazil as a significant part of the analysis.

Table 5. Quality assessment of documents.

Question	Score for the answer		
	Yes	Partially	No
Does the study focus on predatory or abusive litigation?	4	2	0
Does the study focus on AI or legal technology?	2	1	0
Is the study specific to the Brazilian legal system?	2	1	0
Does the study provide empirical evidence or case studies?	2	1	0
Is the methodology well-documented and reproducible?	2	1	0
Are the results and conclusions clearly presented?	2	1	0

Table 6. Data Extraction.

Data
Title of the study or article being reviewed.
Authors of the study.
The year when the study was published.
Methodologies to,ols, or techniques used to identify or combat predatory litigation.
Type of study or research design used.
Key findings or results of the study.
Any limitations noted in the study.
The authors provide suggestions for future research.

- discourage unnecessary litigation and to ensure that judicial access serves those in genuine need.
- **Limitations:** The study highlights that while the hypothetical-deductive method is insightful, additional empirical data would support more robust conclusions about the impact of procedural costs.
 - **Suggestions for Future Research:** Future research could focus on empirical studies to evaluate the effectiveness of proposed reforms in procedural fees and explore alternative models from other countries that could inform sustainable practices in the Brazilian judicial system.
- **Inteligência artificial no sistema jurídico brasileiro: fato ou ficção? (59)**
 - **Authors:** Cristiano Hauck Civitarese
 - **Year:** 2024
 - **Methodology:** Artificial Intelligence, Legal Tech Tools
 - **Design:** Theoretical Analysis
 - **Findings:** AI can streamline judicial operations but faces ethical challenges, including biases in decision-making.
 - **Limitations:** Ethical and legal challenges, such as AI-induced biases.
 - **Suggestions for Future Research:** Ethical guidelines for AI, improved data governance, and standards for AI integration in the judiciary.
 - **O impacto da inteligência artificial no Direito: questões éticas e legais (60)**
 - **Authors:** Gabriel Noll Queiroz, Verônica Silva do Prado Disconzi
 - **Year:** 2024
 - **Methodology:** Artificial Intelligence, Legal Tech Tools
 - **Design:** Theoretical Analysis
 - **Findings:** AI improves efficiency and reduces errors in legal practices but raises concerns about accountability and bias.
 - **Limitations:** AI may lack human nuance, and bias in models can affect fairness.
 - **Suggestions for Future Research:** Ethical frameworks for AI transparency, accountability, and bias mitigation.
 - **Design de sistemas para tratamento da litigância repetitiva (61)**
 - **Authors:** Maria Lúcia Cabral Caruso
 - **Year:** 2023a
 - **Methodology:** Artificial Intelligence, Legal Tech Tools
 - **Design:** Theoretical Analysis
 - **Findings:** Combining mandatory precedent systems with AI improves efficiency in handling repetitive litigation.
 - **Limitations:** Challenges in implementing AI uniformly and the need for consistent data.
 - **Suggestions for Future Research:** Expansion of AI tools for complex cases and harmonizing procedural standards.
 - **Litigância judicial abusiva e instrumentos de gestão processual conferidos ao juiz no código de processo civil: a necessidade de preservação do direito fundamental de acesso à justiça (62)**
 - **Authors:** Joseli Lima Magalhães, Isaac Diego Vieira de Sousa e Silva
 - **Year:** 2024
 - **Methodology:** Legal Tech Tools, Manual Review
 - **Design:** Systematic Review, Theoretical Analysis
 - **Findings:** Brazilian Civil Procedure Code provides tools for controlling abusive litigation while preserving justice access.
 - **Limitations:** Balancing procedural control with constitutional rights.
 - **Suggestions for Future Research:** Technological tools and regulatory approaches for mass litigation.
 - **Jurisdiction and artificial intelligence: reflections and some applications in Brazilian courts (63)**
 - **Authors:** Bruno Makowiecky Salles, Paulo Márcio Cruz
 - **Year:** 2022
 - **Methodology:** Artificial Intelligence, Legal Tech Tools
 - **Design:** Systematic Review
 - **Findings:** AI streamlines judicial processes but requires ethical standards to avoid biases.
 - **Limitations:** Algorithmic bias and need for transparency.
 - **Suggestions for Future Research:** Ethical guidelines for AI, balancing efficiency with fairness.

- **Sistema de justiça 2030: eficácia e racionalidade endereçadas ao usuário e à sociedade (64)**
 - **Authors:** Patrícia Dorigoni Hartmann
 - **Year:** 2022
 - **Methodology:** Other
 - **Design:** Theoretical Analysis
 - **Findings:** The Multi-door approach enhances efficiency in Brazil's judicial system, aligning with the UN's 2030 Agenda goals.
 - **Limitations:** Lacks empirical evidence and detailed implementation guidelines.
 - **Suggestions for Future Research:** Practical implementation of the multi-door system and AI's role within it.
- **TJPA - DIÁRIO DA JUSTIÇA - Edição nº 7510/2022 (65)**
 - **Authors:** Aline Cysneiros Landim Barbosa de Melo, Camila Amado Soares
 - **Year:** 2022
 - **Methodology:** Other
 - **Design:** Other
 - **Findings:** High rates of artificially created cases burden the system, requiring systematic monitoring.
 - **Limitations:** Challenges are harmonizing monitoring efforts across regions.
 - **Suggestions for Future Research:** Enhanced monitoring tools and "Justice 4.0" unit.
- **Julgamento em extinção? The “Vanishing Trial” Eestudo de Marc Galanter e a transformação da atividade jurisdicional no Brasil (66)**
 - **Authors:** Maria Cecília de Araujo Asperti, Susana Henriques da Costa
 - **Year:** 2022
 - **Methodology:** Manual Review
 - **Design:** Theoretical Analysis
 - **Findings:** Trend toward "vanishing trials" in Brazil, with procedural policies reshaping judicial activities.
 - **Limitations:** Lack of comprehensive data in Brazil limits empirical basis.
 - **Suggestions for Future Research:** Judicial management policy impacts justice access.
- **Design de Gestão de Precedentes Qualificados (67)**
 - **Authors:** Alécia Vieira dos Santos
 - **Year:** 2023
 - **Methodology:** Legal Tech Tools
 - **Design:** Empirical Study, Systematic Review
 - **Findings:** Optimizes legal workflow and reduces caseload.
 - **Limitations:** Ongoing training needed.
 - **Suggestions for Future Research:** Data integration and impact analysis.
- **Acesso à Justiça e abuso do direito de ação (68)**
 - **Authors:** João Carlos Leal Junior, Carlos Picchi Neto
 - **Year:** 2016
 - **Methodology:** Manual Review
 - **Design:** Theoretical Analysis
 - **Findings:** Abusive litigation undermines access to justice; mechanisms are needed to prevent abuse.
 - **Limitations:** Measuring abuse objectively is complex.
 - **Suggestions for Future Research:** Procedural reforms to manage abusive practices.
- **Emprego e litigiosidade: impactos da reforma trabalhista na região de Campina Grande - PB (69)**
 - **Authors:** Vanusa Vaniere Nunes Teixeira
 - **Year:** 2022
 - **Methodology:** Other
 - **Design:** Empirical Study
 - **Findings:** Labor Reform decreased cases but led to precarious work conditions.
 - **Limitations:** Not noted in provided excerpts.
 - **Suggestions for Future Research:** Not explicitly mentioned.
- **Entre morosidade e eficiência: estudo sobre justiça e serviço público judiciário no estado brasileiro. (70)**
 - **Authors:** Francisco das Chagas Moreira Neto
 - **Year:** 2023
 - **Methodology:** Manual Review
 - **Design:** Empirical Study
 - **Findings:** Multidisciplinary approach needed to improve judicial service efficiency.
 - **Limitations:** Structural and societal factors affect efficiency.
 - **Suggestions for Future Research:** Inter-institutional approaches to improve judiciary efficiency.
- **Crerios de julgamento por atos de má-fé processual: estudo de caso a partir de julgados do Tribunal de Justiça de Santa Catarina (71)**
 - **Authors:** Eduardo Passold Reis
 - **Year:** 2021
 - **Methodology:** Manual Review
 - **Design:** Case Study, Empirical Study
 - **Findings:** Notes absentee patterns in higher courts for bad faith cases.
 - **Limitations:** Judges' factors affect decisions.
 - **Suggestions for Future Research:** Influence of judge backgrounds on judicial decisions.
- **Incidente de resolução de demandas repetitivas e o princípio da segurança jurídica (72)**
 - **Authors:** Teresa Cristina Alves de Oliveira Viana
 - **Year:** 2024
 - **Methodology:** Manual Review
 - **Design:** Systematic Review, Theoretical Analysis
 - **Findings:** Limited application of IRDR in Maranhão's court system.
 - **Limitations:** Low number of cases processed under this mechanism.
 - **Suggestions for Future Research:** Methods to improve IRDR's adoption in Brazilian courts.

- **Automation mechanisms applied in the legal practice of the Brazilian federal public administration (73)**
 - **Authors:** José Sérgio da Silva Cristóvam, Tatiana Meinhart Hahn
 - **Year:** 2020
 - **Methodology:** Artificial Intelligence, Machine Learning
 - **Design:** Systematic Review
 - **Findings:** AI and automation improve resource allocation and efficiency.
 - **Limitations:** Challenges with data integration.
 - **Suggestions for Future Research:** Expand AI to other federal sectors.
- **Artificial Intelligence and Innovation in Brazilian Justice (74)**
 - **Authors:** Fausto Martin De Sanctis
 - **Year:** 2021
 - **Methodology:** Manual Review
 - **Design:** Systematic Review
 - **Findings:** AI increases speed and efficiency, raising ethical concerns.
 - **Limitations:** Potential for algorithmic biases.
 - **Suggestions for Future Research:** Regulatory measures for bias mitigation.
- **Predicting the number of days in court cases using artificial intelligence (75)**
 - **Authors:** Raphael Souza de Oliveira, Amilton Sales Reis Jr., Erick Giovani Sperandio Nascimento
 - **Year:** 2022
 - **Methodology:** Artificial Intelligence, Machine Learning
 - **Design:** Empirical Study
 - **Findings:** Adaboost model predicts case durations with 84% accuracy.
 - **Limitations:** Imbalance in the dataset affects model performance.
 - **Suggestions for Future Research:** Dataset balancing and feature relevance analysis.
- **Revolution 4.0 in Judicial Power: Mapping the Use of Artificial Intelligence in Brazilian Courts (76)**
 - **Authors:** Fernanda Bragança, Laurinda Fátima da F. P. G. Bragança
 - **Year:** 2019
 - **Methodology:** Artificial Intelligence
 - **Design:** Systematic Review, Theoretical Analysis
 - **Findings:** AI adoption increases case processing speed; highlights ethical concerns.
 - **Limitations:** Rapid evolution of AI technologies and transparency issues.
 - **Suggestions for Future Research:** Ethical AI usage and effectiveness evaluation.
- **Using deep learning to predict outcomes of legal appeals better than human experts (77)**
 - **Authors:** Elias Jacob de Menezes-Neto, Marco Bruno Miranda Clementino
 - **Year:** 2022
 - **Methodology:** Artificial Intelligence, Machine Learning
 - **Design:** Case Study, Empirical Study
 - **Findings:** Deep learning models outperform experts in predicting legal outcomes.
 - **Limitations:** Solely relies on first-instance court decision texts.
 - **Suggestions for Future Research:** Include additional data sources to improve accuracy.
- **Predicting Brazilian court decisions (78)**
 - **Authors:** André Lage-Freitas, Héctor Allende-Cid, Orivaldo Santana, Livia Oliveira-Lage
 - **Year:** 2022
 - **Methodology:** Machine Learning
 - **Design:** Empirical Study
 - **Findings:** BERT and XGBoost models effectively predict Brazilian court outcomes.
 - **Limitations:** Dataset size limits model performance.
 - **Suggestions for Future Research:** Customizing datasets by judge, law subject, and court level.
- **Tecnologia da informação como aliada no combate ao fenômeno da explosão de litigiosidade: busca e tratamento de dados, identificação, diagnóstico e limitações (79)**
 - **Author:** Rodrigo Martins Faria
 - **Year:** 2023
 - **Methodology:** Manual Review
 - **Design:** Theoretical Analysis
 - **Findings:** In the context of explosion of litigation, the multiplication of repetitive demands puts the capacity of the Judiciary to fulfill its constitutional mission of social pacification in check. The 4.0 concept refers to the fourth and current Industrial Revolution, driven by the paradigm of digital transformation and disruptive technologies, such as artificial intelligence, as a scenario of the technological revolution which the judiciary has been going through, in order to adapt to the new social context. Some tools developed and in production by Brazilian courts were presented, which represent the breaking of a new frontier towards the use of technology as an ally of the Judiciary in identifying and in the diagnosis of litigation.
 - **Limitations:** The idea of a Judiciary 4.0, driven by data and intelligence artificial, is facing a real movement of resistance by a large part of the legal community in regarding the use of artificial intelligence in the Judiciary.
 - **Suggestions for Future Research:** The development of technological tools that allow the courts to manage the immense collection of repetitive actions contributes decisive way to face the problem of slowness and inefficiency of the justice system.
- **Litigância predatória: entre o acesso à justiça e os abusos sistemáticos do direito ao processo (80)**

- **Authors:** Marco Bruno Miranda Clementino and Lucas José Bezerra Pinto
 - **Year:** 2024
 - **Methodology:** Manual Review
 - **Design:** Theoretical Analysis
 - **Findings:** This article analyzes the main elements that represent the idea of predatory litigation and its distinctions from other similar types of procedural abuse, in a context of collision with the right of access to justice. To this end, a problematizing and exploratory approach is adopted, without giving up some deductive findings. It aims to build a institutional and procedural format necessary to consolidate steps (identification, proceduralization and accountability) to address this abusive phenomenon.
 - **Limitations:** This abusive phenomenon has been detected at the jurisdictional level without further methodological digressions.
 - **Suggestions for Future Research:** It reveals a collectivizing conception immanent to predatory litigation, attracting the use of systemic mechanisms to respond to systemic problems, such as cooperative institutionalization (cooperation between different actors in the justice system to administer it) and integrative collectivization (procedural model of communicability between different collective-repetitive-cooperative procedural techniques).
- **O tratamento das novas faces da litigiosidade: das espécies anômalas à litigância predatória (81)**
 - **Author:** Taís Schilling Ferraz
 - **Year:** 2024
 - **Methodology:** Manual Review
 - **Design:** Theoretical Analysis
 - **Findings:** The litigiousness phenomenon, traditionally represented by a concentration of repetitive demands entering the Judiciary, has recently evolved, exhibiting new manifestations, including predatory behaviours or demands which bring, in their wake, reckless, frivolous conduct and even fraud. The article aims to distinguish specific anomalous manifestations of litigiousness and explore their perception and treatment. Utilizing bibliographical, documentary, and database research, it presents a comprehensive overview of litigiousness, situating the topic within the context of the abuse of rights in civil procedure, identifying the particularities of anomalous conduct or demands and ultimately defining the elements of predatory litigation.
 - **Limitations:** In the face of these new manifestations of litigiousness and within the assumptions of systemic thinking, the article assesses the (in)effectiveness of traditional strategies for treating conflicts, based on the constant pursuit of increased efficiency.
 - **Suggestions for Future Research:** The article proposes a structural and collaborative approach to the phenomenon, drawing on the work conducted by the intelligence centers of the judiciary and centers for monitoring demand profiles.

Technical Review

This section provides a technical review of recent initiatives by Brazilian courts that employ artificial intelligence to improve judicial efficiency and counteract predatory litigation. The following list presents each initiative, detailing the court, title, brief description, objective, approach, and if there is a focus of the litigation (82; 83).

- **Court:** Tribunal Regional do Trabalho da 4^o Região
Title: Artificial Intelligence Conciliability Index - ICia
Brief Description: The AI Conciliability Index is a metric that calculates the likelihood of a case being resolved through conciliation, using historical data to predict outcomes.
Objective: To improve judicial processes by applying AI to improve case resolution efficiency and support alternative dispute mechanisms.
Approach: Employs AI techniques such as machine learning models, data analytics, and automation to enhance decision-making and case management.
Litigation Focus: No specific focus on repetitive or predatory litigation.
- **Court:** Tribunal Regional do Trabalho da 13^a Região
Title: AI-Driven Judicial Startups: PJe ((Electronic Judicial Process) Admissibility, PJe Correction, and Automatic Procedural Summary
Brief Description: These AI-powered startups focus on automating specific judicial processes within TRT13, including verifying appeal requirements, continuous correctional activity to detect procedural inconsistencies and automated summarization of case statuses.
Objective: To improve efficiency, accuracy, and accessibility in judicial processes by reducing manual effort and enhancing process transparency and speed.
Approach: Utilizes AI and automation to perform machine-based verification, correctional monitoring, and automated summarization of case statuses, leveraging data analytics and real-time notifications to streamline judicial workflows.
Litigation Focus: Focus on improving procedural efficiency and consistency without a specific focus on repetitive or predatory litigation.
- **Court:** Tribunal Regional Eleitoral de Pernambuco
Title: Automation of Electronic Voting Machine Audit with AI
Brief Description: This initiative aims to automate the audit of electronic voting machines using robotics and artificial intelligence, ensuring accuracy and transparency in the Brazilian electoral process by minimizing human intervention.

Objective: To enhance the efficiency and credibility of election audits by replacing human efforts with AI-driven robotic solutions, ensuring compliance with TSE regulations and improving public trust.

Approach: Employs AI techniques such as visual recognition, robotic programming, and automation of audit procedures to enable robots to perform tasks traditionally done by human auditors, including verifying candidate numbers and recording votes accurately.

Litigation Focus: No specific focus on litigation; primarily concerned with process transparency and reliability in election audits.

- **Court:** Tribunal de Justiça do Estado do Rio Grande do Norte

Title: GPSMED - Analysis of Judicialization in the Health Area

Brief Description: GPSMED is a data platform developed with AI techniques to collect procedural information related to public health, aimed at assisting public administration in resource management and supporting the judiciary in reducing healthcare litigation.

Objective: To support judicial and public health administration by improving resource allocation, streamlining judicial processes, and providing insights for strategic decision-making in public health.

Approach: Utilizes AI through data collection, analysis, and visualization in dashboards to aid in decision-making for public health management and judicial processes.

Litigation Focus: Focus on healthcare-related cases to facilitate resource planning and reduce healthcare litigation.

- **Court:** Tribunal de Justiça de Pernambuco

Title: Bastião

Brief Description: The Bastião Project is a system that identifies and addresses predatory litigation using data analysis and AI to classify potentially harmful legal demands early in the judicial process.

Objective: To improve judicial efficiency by identifying and managing predatory or repetitive demands in the court system, thus reducing case congestion and average processing time.

Approach: Utilizes AI models, including neural networks, decision trees, and regression algorithms, in conjunction with a database of structured attributes. It includes a consultative interface and tagging system for potential predatory cases.

Litigation Focus: Focused on identifying repetitive and predatory litigation early in the filing process.

- **Court:** Tribunal de Justiça do Piauí

Title: JuLIA - Artificial Intelligence-Assisted Justice

Brief Description: The JuLIA (in Portuguese: Justiça Auxiliada por Inteligência Artificial) project utilizes Artificial Intelligence to automate routine judicial tasks such as process classification, detection of duplicate cases, and drafting of judicial decision

drafts. It also enhances information accessibility for the public and supports communication with citizens and legal representatives.

Objective: To modernize judicial processes using AI for improved efficiency and accessibility and to address the challenges of increasing case volumes.

Approach: Uses AI for data analysis, automated messaging through platforms like WhatsApp, chatbot services, and predictive analytics for case demand forecasting to reduce decision subjectivity and improve information management.

Litigation Focus: General application with benefits for accessibility and efficiency, without a specific focus on repetitive or predatory litigation.

- **Court:** Tribunal de Justiça do Estado de Goiás

Title: - Berna - Electronic Search in Records using Natural Language

Brief Description: Berna (in Portuguese: Busca Eletrônica em Registros usando Linguagem Natural) is a solution designed to identify and automatically unify large volumes of judicial demands that share similar facts and legal theses in the initial petitions, enhancing the efficiency and speed of judicial services.

Objective: To expedite the judicial process by automating case grouping and improving the delivery of judicial services through AI.

Approach: Utilizes AI techniques for natural language processing to read initial petitions, categorize requests and legal theses, and form groups of similar cases with a security coefficient of around 90%
Litigation Focus: Focuses on identifying and managing repetitive judicial demands effectively.

- **Court:** Tribunal de Justiça do Maranhão

Title: Sentença + Simples (84)

Brief Description: Simplified Judgement initiative is a pilot to streamline the judgement process for faster, accessible resolutions.

Objective: To improve judicial processes by applying AI to improve case resolution efficiency and support alternative dispute mechanisms.

Approach: Employs AI techniques such as machine learning models, data analytics, and automation to enhance decision-making and case management.

Litigation Focus: No specific focus on repetitive or predatory litigation.

- **Court:** Tribunal de Justiça do Ceará

Title: Campaign to Combat Domestic Violence with AI Multilingual Support

Brief Description: This campaign employs artificial intelligence to educate tourists about domestic violence laws in Brazil through multilingual content. An AI avatar named "Ju" delivers information in six languages, enhancing accessibility and awareness.

Objective: To raise awareness among international tourists about the Brazilian domestic violence law and available reporting channels, aiming to prevent and combat violence against women.

Approach: Utilizes AI-driven avatars and QR codes linking to educational videos in multiple languages, enabling effective communication and outreach to diverse audiences.

Litigation Focus: Primarily focused on public awareness and prevention, without specific litigation focus.

- **Court:** Tribunal de Justiça do Estado de Goiás

Title: #Simplificar 5.0

Brief Description: The #Simplificar 5.0 Project uses AI to create illustrated summaries of judicial sentences sent via messaging apps to the involved parties and lawyers, employing simple language and visual law to enhance understanding.

Objective: To improve access to justice by simplifying judicial language and incorporating visual design and AI, enabling citizens to fully comprehend legal decisions.

Approach: The project uses a machine learning algorithm to classify sentences, draft illustrated summaries with human validation, and deliver these summaries through messaging apps, focusing on simplified language and visual representation.

Litigation Focus: General application to facilitate understanding of judicial decisions, with no specific emphasis on repetitive or predatory litigation.

- **Court:** Tribunal de Justiça do Distrito Federal e Territórios

Title: SAREF - Remote Presentation and Facial Recognition System

Brief Description: The SAREF (in Portuguese: Sistema de Apresentação Remota e Reconhecimento Facial) is an AI-based project designed to securely facilitate the periodic presentation of individuals serving sentences under an open regime, a process challenged by the Covid-19 pandemic. It uses facial recognition and remote monitoring to ensure compliance.

Objective: To enable safe and accessible criminal presentations remotely, supporting the judicial process while accommodating accessibility and usability needs.

Approach: Utilizes AI techniques in facial recognition and user data analytics, along with mobile and self-service kiosk versions for remote and in-person access. Emphasis is placed on accessibility and user-friendly design.

Litigation Focus: Criminal justice focuses on individuals with restricted movement.

- **Court:** Tribunal Regional do Trabalho da 4ª Região

Title: SINARA

Brief Description: SINARA is an AI algorithm designed for extracting legal information from texts, particularly normative provisions and precedents. It aids in understanding legal texts by identifying common legal grounds across cases.

Objective: To enhance legal processes by using AI to rank and analyze cases, facilitating more efficient

legal information extraction and decision-making.

Approach: Utilizes Natural Language Processing (NLP) techniques to extract legal provisions accurately, maintaining high accuracy over time due to linguistic rather than purely legal dependencies.

Litigation Focus: No specific focus on repetitive or predatory litigation.

- **Court:** Justiça Federal da 3ª Região

Title: SIGMA

Brief Description: SIGMA is a centralized system that uses artificial intelligence to rank and recommend templates by identifying key information in case files, thus suggesting models previously used in similar cases.

Objective: To streamline judicial drafting processes by employing AI to improve efficiency and consistency across case documentation.

Approach: Utilizes natural language processing (NLP) and AI-based ranking to analyze legal documents, recommending suitable templates to enhance decision-making.

Litigation Focus: No specific mention of a focus on repetitive or predatory litigation.

- **Court:** Tribunal de Justiça do Maranhão

Title: Qualified Precedent Panel

Brief Description: The Qualified Precedent Panel is an AI-driven tool that helps judges identify relevant precedents in similar cases, streamlining judicial decision-making by providing quick access to applicable past rulings.

Objective: To enhance judicial efficiency and speed up case resolution by utilizing AI to identify applicable precedents in repetitive litigation.

Approach: Utilizes AI techniques to analyze text similarities in case documents, highlighting probable precedents from a database and facilitating access to recurring precedents based on location, case type, and other filters.

Litigation Focus: Primarily aimed at repetitive litigation cases, enabling faster resolution by referencing established precedents.

- **Court:** Tribunal de Justiça do Paraná

Title: Tecnologia de Reconhecimento de Voz para Assistente Virtual

Brief Description: The voice recognition technology enables users to interact with the justice system through speech or typing, facilitating access to judicial information services, especially for those with difficulty with conventional input devices.

Objective: To improve digital inclusion and judicial accessibility by leveraging voice recognition for easier access to judicial services.

Approach: Implements AI-driven voice recognition to enhance user interaction in judicial services, providing alternative communication methods for accessibility.

Litigation Focus: Not specifically targeted toward any litigation focus, but enhances accessibility across

general judicial services.

- Court:** Tribunal de Justiça do Paraná
Title: SMART JUSTICE:
Brief Description: The SMART JUSTICE initiative aims to innovate productivity analysis and operational capacity by leveraging AI and data analytics to integrate various information sources, including staff tenure, leave history, work calendar, internal activities, team size, case history, and even regional weather conditions, to estimate productivity with high precision and aid in management.
Objective: To enhance judicial productivity management and optimize resource allocation through AI and advanced analytics.
Approach: Uses AI-driven analytics and data integration to provide accurate productivity estimates, focusing on comprehensive data correlations.
Litigation Focus: No specific focus on repetitive or predatory litigation.
- Court:** Tribunal de Justiça do Paraná
Title: LARRY System for Document Similarity Identification
Brief Description: The LARRY system utilizes artificial intelligence to identify similarities between legal documents, aiding in identifying repetitive or mass claims.
Objective: To enhance process speed and reduce case backlog in civil and criminal matters, supporting the judiciary's goals by suggesting solutions and improvements.
Approach: Implements AI and automation algorithms to assist various branches of the TJPR in detecting repetitive demands, thereby improving process efficiency and productivity.
Litigation Focus: Focus on identifying and addressing repetitive or mass litigation cases.
- Court:** Tribunal de Justiça do Ceará
Title: Simplified Legal Language Model Implementation
Brief Description: The initiative introduces a simplified language model in legal documents, making information more accessible and understandable for the general public. A QR Code with an AI-generated video explanation is included to aid comprehension.
Objective: To improve communication between the judiciary and society by simplifying legal language, facilitating access to justice, and promoting inclusivity.
Approach: Utilizes elements of "Visual Law" and AI for video explanations, aiming to transform legal information into a format accessible to all.
Litigation Focus: Emphasis on public awareness and access to justice rather than specific types of litigation.
- Court:** Tribunal de Justiça do Rio Grande do Norte
Title: Simplification of Precedents
Brief Description: This project aims to simplify the understanding of judicial precedents established by

the court, particularly in repetitive demand resolution incidents and similar cases, to make legal theses more accessible and clear for broader audiences.

Objective: To enhance the accessibility and clarity of legal precedents, aligning with sustainable development goals and supporting legal understanding for all stakeholders.

Approach: Simplifies legal texts using the FIRAC(Facts-Issue-Rule-Application/Analysis-Conclusion) model to break down complex cases into structured components, and aligns each precedent with the UN's 2030 Sustainable Development Goals.

Litigation Focus: Selected repetitive demand cases, focusing on the applicability of tax exemptions in environmental preservation areas.

- Court:** Tribunal de Justiça do Estado da Bahia
Title: Oxóssi
Brief Description: The Oxóssi system is an AI-driven tool that enhances the efficiency of judicial process management by allowing intelligent keyword searches and similarity-based document analysis, assisting judges and staff in handling the judicial docket.
Objective: To improve process management and productivity within the court by providing advanced tools for locating and analyzing judicial documents.
Approach: Combines intelligent text searches, similarity-based document matching, and advanced filtering options with a robust data lake infrastructure, enabling comprehensive case management and automation of routine tasks.
Litigation Focus: General case management focuses on identifying mass or predatory litigation and repetitive legal themes.
- Court:** Tribunal de Justiça de Pernambuco
Title: TJ Colab - Collaborative Platform with AI
Brief Description: TJ Colab is a collaborative platform that leverages AI to facilitate access to and sharing of information via indexed repositories, aiming to improve workflow for court employees.
Objective: To create a reliable and accessible collaborative environment where employees can easily find and share resources and solutions, enhancing work efficiency and decision-making.
Approach: Utilizes AI to index, organize, and continuously improve the repository of documents, tutorials, and manuals, supporting secure and streamlined access to relevant information.
Litigation Focus: Broad focus on various legal specializations, with resources for civil, criminal, and other court departments.
- Court:** Tribunal de Justiça de Pernambuco
Title: Atena AI System
Brief Description: The Atena AI System is a humanized artificial intelligence platform designed to support PJe system users. It aims to improve data entry accuracy and facilitate user support through AI-driven responses and suggestions while tracking frequent inquiries for better process management.

Objective: To enhance procedural accuracy and speed by reducing data entry errors and providing detailed user guidance, thus improving judicial efficiency.

Approach: Uses AI to provide real-time responses, log frequent questions, and maintain a database of user inquiries to inform training and process improvements.

Litigation Focus: Focus on reducing procedural errors in judicial data entry, without emphasis on repetitive or predatory litigation.

- **Court:** Tribunal de Justiça de Pernambuco

Title: JusIA

Brief Description: JusIA is a platform designed to structure, classify, and enhance objectivity in the judicial complaint process, transforming basic user information into structured, standardized petitions, which aim to reduce subjectivity and improve accessibility.

Objective: To streamline judicial procedures, increase efficiency, and make legal processes more accessible to the public through automated and structured data processing.

Approach: Uses AI-driven data structuring and classification, enabling quick conversion of oral or unstructured complaints into formalized petitions, providing a user-friendly response in plain language.

Litigation Focus: Focuses on processing mass demands effectively by providing structured classification and indexing for faster resolutions and better access to justice.

- **Court:** Tribunal de Justiça de Pernambuco

Title: DORA

Brief Description: DORA is an AI-based system for data validation in judicial processes. It analyzes and suggests corrections, identifies essential missing fields, and provides conformity indexes, aiming to enhance data accuracy and reduce process errors.

Objective: To streamline judicial processes by improving data integrity and reducing inconsistencies that delay case processing times.

Approach: Uses AI for data validation, suggesting fields and classifications based on a semantic analysis of case data. It supports decision-making and optimizes workflow through automation and interactive virtual engagement.

Litigation Focus: Primarily aimed at improving efficiency in data handling within the judiciary, not specifically targeting repetitive or predatory litigation.

- **Court:** Tribunal de Justiça de Pernambuco

Title: Connect Rock

Brief Description: The project leverages AI to improve case processing efficiency by identifying bottlenecks and prioritizing critical cases. It integrates with the PJe system, providing an agile, automated approach to managing caseloads effectively.

Objective: To enhance judicial workflow and productivity by using AI for task prioritization and management, ultimately reducing processing times

and improving service quality for court staff.

Approach: Applies AI-driven analytics to automate identifying high-priority cases, streamline workflows, and support staff with task allocation based on real-time data and process urgency. Uses agile methodologies such as SCRUM (85) alongside AI to manage workload distribution.

Litigation Focus: Targets optimization in case processing without a specific focus on any litigation, aiming instead at overall efficiency and case resolution improvements.

- **Court:** Tribunal de Justiça de Pernambuco

Title: Anjoário - Virtual Angel of Justice

Brief Description: The Virtual Angel is an AI-driven virtual assistant designed to address user issues related to court processes, responding to repetitive demands through an integrated WhatsApp chatbot system. It aims to streamline communication and reduce the workload of judicial staff.

Objective: To provide 24/7 support to citizens, reduce delays in judicial response and alleviate workload by automating responses to frequently asked questions.

Approach: Utilizes AI technologies such as a database of frequently asked questions, machine learning for predictive responses, and integration with various TJPE systems to enhance user access and case management.

Litigation Focus: Primarily designed to handle repetitive procedural queries but not specifically focused on content-specific repetitive or predatory litigation.

- **Court:** Tribunal de Justiça de Pernambuco

Title: Justec Bot - Automated Summons and Notifications System

Brief Description: Justec Bot is an automated platform designed to expedite the process of issuing and sending court summonses and notifications. It integrates with the PJe system to generate documents based on templates, reducing manual tasks for court staff.

Objective: To improve the efficiency and speed of court administrative processes by reducing repetitive tasks for court staff, allowing them to focus on substantive case analysis.

Approach: Utilizes AI and automation to streamline citation and notification processes, with planned future extensions for broader document management within court procedures.

Litigation Focus: Initially aimed at large litigants, with potential long-term applications for smaller cases as digital adoption expands.

- **Court:** Tribunal de Justiça de Pernambuco (TJPE)

Title: Genesis - AI Solution for Identifying Aggressive and Repetitive Litigations

Brief Description: Genesis is an AI-driven solution designed to identify potentially predatory legal demands by analyzing a large database of cases. The system classifies and isolates repetitive or aggressive demands, supporting the judicial process

with automated notifications for verification and authentication.

Objective: To improve the efficiency of judicial case management, reduce predatory litigations, and provide faster access to justice for the public.

Approach: Utilizes artificial intelligence, including machine learning, metadata analysis, and automated data processing, to assess the likelihood of repetitive or aggressive litigation patterns and manage these cases effectively.

Litigation Focus: Primarily targets predatory and repetitive litigation patterns to prevent abuse of judicial resources.

- **Court:** Tribunal de Justiça de Pernambuco

Title: Sherlock - Predatory Demand Identification System

Brief Description: Sherlock is an AI-based RPA (Robotic Process Automation) designed to detect predatory demands when initiating a legal case. It flags these demands with a popup notification and channels them through a specialized procedural flow, parallel to the PJe system.

Objective: To reduce judicial backlog and increase procedural efficiency by identifying and standardizing the handling of predatory lawsuits through automated case categorization.

Approach: Utilizes AI techniques, including data aggregation, pattern recognition, and automated decision-making, to detect and manage cases deemed predatory. It systematically flags similar cases based on specific parameters, such as repetitive claims or lack of substantial evidence, enabling automated routing and decision-making processes.

Litigation Focus: Primarily targets mass litigation and repetitive lawsuits against entities like financial institutions, telecom companies, and utility providers.

- **Court:** Tribunal de Justiça de Pernambuco

Title: Blind - A System for Predatory Litigation Detection

Brief Description: The Blind system is an AI-driven solution designed to protect the judiciary from predatory litigation practices. It analyzes case patterns and uses natural language processing to detect potential fraudulent cases, preventing undue strain on the judicial system.

Objective: To safeguard the judiciary from fraudulent mass litigation by filtering out predatory cases at the initial stages and preventing them from proceeding through the system.

Approach: Utilizes AI and automation techniques, including natural language processing, criteria-based filtering, and automated alerts, to identify red flags like repetitive language, high case volume from specific attorneys, and inconsistencies in documentation.

Litigation Focus: Predatory litigation, particularly cases with high volumes, fraud indicators, and cases involving repetitive or falsified documentation.

- **Court:** Tribunal de Justiça do Rio Grande do Norte

Title: Combating Predatory Litigation: Innovative

Strategies to Promote Efficiency in the Judiciary of Rio Grande do Norte

Brief Description: The initiative aims to combat predatory litigation by leveraging innovation and strategic methodologies, such as Design Thinking, to address public challenges and improve efficiency within the judicial system.

Objective: To reduce the volume of predatory lawsuits, which exploit the system to cause delays and additional costs, thereby enhancing the efficiency of the judiciary.

Approach: Utilizes data analytics and is evolving towards incorporating AI and predictive analysis tools. AI will assist in identifying patterns of predatory litigation, aiding case management and decision-making processes.

Litigation Focus: Specifically targets repetitive and predatory litigation, aiming to detect and mitigate cases that misuse the judicial process.

- **Court:** Tribunal Regional do Trabalho da 17ª Região

Title: Conciliômetro - Statistical Models for Conciliation

Brief Description: The Conciliômetro project involves creating statistical models to predict conciliation likelihood in judicial processes. It supports units involved in conciliation activities by aiding in process selection.

Objective: To increase the conciliation rate within the court and support the strategic goals of ensuring adequate treatment of labour conflicts and reasonable case duration.

Approach: Uses statistical modelling and data analysis on judicial cases to estimate conciliation probabilities, with visual dashboards displaying categorized processes based on their likelihood of conciliation.

Litigation Focus: Focuses on labour conflict resolution and conciliation to reduce case backlog.

- **Court:** Tribunal Regional Eleitoral de Pernambuco

Title: Disinformation Combat Project

Brief Description: The Disinformation Combat Project aims to proactively identify and address misinformation related to Electoral Justice on social media by using AI tools to monitor and analyze social media content.

Objective: To enhance the integrity of electoral processes by identifying and clarifying misinformation through AI-driven analysis and monitoring of social media.

Approach: Utilizes Design Thinking principles, partnership with the Universidade Federal de Pernambuco, and AI techniques, including Machine Learning and Natural Language Processing, to monitor, classify, and respond to social media content.

Litigation Focus: No specific focus on repetitive or predatory litigation.

- **Court:** Tribunal de Justiça da Bahia

Title: FABBOT Project - Robot Factory

Brief Description: The FABBOT Project represents an innovative milestone, integrating advanced technology and legal expertise to optimize judicial functions in both Special Courts and Common Justice. The core of this initiative is a robust framework that allows for scalable robot creation that is compatible with most browsers and supports various judicial tasks.

Objective: To enhance judicial efficiency by automating repetitive tasks, allowing legal professionals to focus on analytical responsibilities and ensuring greater accuracy.

Approach: Utilizes a scalable Framework with AI capabilities for tagging, filtering, and task automation, deploying over 50 robots in systems like PJe and PROJUDI to execute functions like dispatching notifications, case reassignment, thematic triage, and document preparation.

Litigation Focus: Focus on optimizing judicial processes by reducing repetitive and time-consuming tasks.

- **Court:** Tribunal de Justiça do Amazonas (86; 87; 88)

Title: Arandu/Abaçai

Brief Description: The Arandu tool was designed to identify repetitive and predatory demands in the judicial system. The Arandu artificial intelligence system has the ability to analyze the court's collection in real time, making it possible to verify the similarity of a current case with the court's complete history. The Arandu AI model was trained based on more than 500.000 complaints distributed in the local judiciary, and is capable of learning from each new complaints launched within the system. Today, the tool's learning base already has around 700.000 complaints in a system that is constantly updated. The tool is available to use in two ways. The first is integrated into the system, in which the worker can check the similarity of the process under analysis with those already learned by the AI. The second is through a panel in which the worker can carry out broader searches, for example "District", "Court", "Party Name" and "Process Number" The panel includes the e-SAJ and Projudi systems.

Objective: Aims to speed up judicial procedures. When receiving a new complaint or when consulting an existing one, the system compares those documents in real time with the court's complete history.

Approach: The main benefits include: the identification of predatory and repetitive lawsuits; automation of the process of comparing cases, freeing up time for judges and workers for other tasks and; reduction of human resources costs and procedural processing time.

Litigation Focus: Focused on identifying repetitive and predatory litigation.

- **Court:** Tribunal de Justiça do Distrito Federal e Territórios (86; 89)

Title: Ártemis

Brief Description: Artémis was developed to track

and analyze all lawsuits in the PJe, within a few minutes of being filed with the TJDFT, with the aim of identifying cases of *lis pendens*^{*}, prevention or *res judicata*[†].

Objective: Identify cases of *lis pendens*, prevention or *res judicata*.

Approach: If the tool identifies a possible case of *lis pendens*, prevention or *res judicata*, it will signal the process in the PJe with a label and a specific flow task will be made available for due treatment in the responsible sector.

Litigation Focus: Focus on identifying cases of *lis pendens*, prevention or *res judicata*, no specific focus on predatory litigation.

Discussion

This section synthesizes the insights from both scientific literature and technological tools addressing repetitive demands and abusive litigation in the Brazilian judiciary. Efforts to curb abusive litigation are evident in various courts, with a range of methodologies, from legal analyses to AI-based tools aimed at detecting and managing repetitive claims.

Scientific Literature on Abusive Litigation

The literature review highlights a growing concern about repetitive litigation, often termed predatory or abusive, which places considerable strain on the judiciary. Brito (2024) found that the measures of the Special Court Intelligence Centre in Rio Grande do Norte significantly reduced cases of undue inclusions in the debtor registry, suggesting an initial success in deterring abusive claims through administrative oversight (57). Similarly, Souza et al. (2022) argued for a procedural cost reform to discourage frivolous lawsuits, proposing that equitable access to justice requires disincentives for unwarranted claims (58). Other studies, such as those of Magalhes and Silva (2024), analysed the tools of the Brazilian Civil Procedure Code to balance the control of abusive litigation while maintaining access to justice, highlighting the need for regulatory improvements (62).

According to Viaro (2023) (15), thinking about the problem of predatory litigation also involves reflecting on the conditions that allow its existence. The Brazilian system establishes exaggerated subsidies, either due to the low value of costs or the high percentage of cases in which the free legal aid benefit is granted, which creates incentives for filing lawsuits. The results are increased costs, congestion and delays. Paradoxically, the excess of access to justice generates the denial of access to justice, making it necessary to search for mechanisms capable of recalibrating the system.

Clementino and Pinto (2024) (80) analyzed the fundamental elements that represent the idea of predatory litigation and its distinctions from other similar types

*A *lis pendens* is a written notice that a lawsuit has been filed concerning real estate.

†a matter that has been adjudicated by a competent court and may not be pursued further by the same parties.

of procedural abuse in order to build the institutional and procedural format necessary to consolidate steps (identification, proceduralization and accountability) to address this abusive phenomenon, which can effectively impact the administration of justice.

Ferraz (2024) (81) proposed a structural and collaborative approach to the phenomenon, drawing on the work conducted by the intelligence centers of the judiciary and monitoring centers of demand profiles, due to the (in)effectiveness of traditional strategies for treating conflicts, based on the constant pursuit of increased efficiency.

In the context of explosion of litigation, Faria (2023) (79) highlighted some tools developed and in production by Brazilian courts, which represent the breaking of a new frontier towards the use of technology as an ally of the Judiciary in the diagnosis of litigation.

Technological Tools for Managing Repetitive Litigation

From a technical perspective, Brazilian courts have adopted AI tools to streamline the detection and management of repetitive claims. For example, the Bastião Project from the Tribunal de Justiça de Pernambuco (TJPE) identifies potentially predatory cases using machine learning algorithms that classify harmful legal demands early in the process (83). Another significant initiative, the Genesis system, uses AI to isolate repetitive or aggressive demands, preventing system abuse and promoting justice efficiency (82). Furthermore, the Sherlock system uses AI for real-time detection of mass claims, flagging cases that exhibit characteristics of predatory litigation, particularly against entities like financial and telecom companies.

In addition, "Bliind" prototype uses AI to check whether the plaintiff's complaint has similarity to another already filed. The system also allows you to check whether the lawyer has an above-average distribution of cases, especially when from a different state of the federation. Furthermore, it also allows the identification of proof of residence (date and name), as well as verifying whether the power of attorney was signed by the party. If a high degree of predatory indication is identified, the party's personal appearance is suggested and the process is forwarded to a specific folder, in the event of non-attendance, a draft terminative decision is prepared (90).

The Bastião Project was developed based on the solutions created in the macro challenge "Aggressive or predatory demands of the system", uniting the three innovation prototypes mentioned above (Bliind, Sherlock and Gênesis), from the "JESP JAM 2022" (an open innovation event promoted by the laboratory of innovation of the Pernambuco Court of Justice) (91; 79).

The Tribunal de Justiça do Amazonas (TJAM) has developed the Arandu/Abaçai tool, which employs artificial intelligence to identify repetitive and predatory litigation. By analyzing cases in real time and comparing them to historical data, the tool aims to improve judicial efficiency and accelerate case handling. Arandu automates similarity detection and analysis, reducing workload and procedural delays while targeting repetitive and predatory cases.

The Tribunal de Justiça do Distrito Federal e Territórios (TJDFT) introduced Ártemis, an AI tool designed to track and analyze lawsuits for cases of *lis pendens*, prevention, or *res judicata*. Integrated with judicial systems, Ártemis identifies and flags problematic cases, streamlining the judicial process by reducing redundancies and procedural inefficiencies. Although its focus is not explicitly on predatory litigation, Ártemis significantly enhances case management efficiency.

Table 7 provides a summary of the primary research papers and technological tools addressing repetitive and predatory litigation in the Brazilian judiciary.

The integration of AI in these tools reflects a proactive approach to handling abusive litigation, aligning with suggestions from the literature to reduce judicial congestion and maintain fairness. Studies and tools indicate a shared goal: ensuring judicial accessibility while safeguarding resources. Continued exploration of AI ethics, as suggested by (59) Civitarese (2024), and refinement of fee regulations may further enhance these efforts, ultimately supporting a fair and efficient judiciary in Brazil.

About research questions

To answer the overarching research question, "What are the current methods and tools used to identify predatory or abusive litigation lawsuits in Brazil?" we address each specific research question (SRQ) based on findings from the literature and technological reviews.

SRQ1: How has artificial intelligence been applied to combat or mitigate predatory litigation in legal systems, particularly Brazil? AI applications in Brazil's judiciary focus primarily on classifying and identifying abusive litigation patterns. Systems such as the "Genesis" and "Sherlock" use machine learning algorithms to detect repetitive and aggressive legal demands, providing early classification and flagging for judicial attention. These tools reflect a trend towards automating identification and pre-emptive case management, aiming to reduce system backlog and enhance resource efficiency.

SRQ2: What machine learning approaches have proven effective in identifying patterns of abusive litigation in lawsuits? Effective machine learning techniques include natural language processing for text similarity analysis and classification algorithms for detecting repetitive filings. For example, the "Sherlock" system leverages NLP to detect patterns of predatory litigation, specifically in repetitive claims within consumer finance and telecom cases. Similarly, the "Bastião" project utilises decision tree and regression algorithms, which are crucial in filtering and tagging potentially predatory claims.

SRQ3: How do AI-based methods compare to traditional human analysis and manual case review in detecting predatory litigation? Compared to manual reviews, AI methods enhance processing speed and accuracy. For instance, the "Blind" system's NLP capabilities significantly outperform manual screening by swiftly identifying fraudulent and repetitive cases that would otherwise overwhelm judicial staff. Additionally, AI-driven systems, like those used in the "Genesis" project, automate

Table 7. Summary of Research Papers and Technological Tools on Repetitive and Predatory Litigation in Brazil

Research/Tool	Key Findings/Objective
Enfrentamento da Litigância Predatória (57)	Implementation of measures to deter predatory litigation in Rio Grande do Norte
Acesso Inautêntico à Justiça (58)	Advocates procedural cost reform to curb frivolous litigation
Litigância judicial abusiva (62)	Examines tools in the Civil Procedure Code to manage abusive litigation
Tecnologia da Informação e Litigiosidade (79)	Highlights technological tools developed to streamline judicial processes and address the explosion of litigation
Predatory Litigation Analysis (80)	Proposes institutional frameworks to address predatory practices while preserving access to justice
New Faces of Litigiousness (81)	Examines anomalous litigation behaviors and proposes systemic, collaborative approaches for resolution
Systemic Conditions of Predatory Litigation (15)	Reflects on subsidies and free legal aid as systemic enablers of predatory litigation, calling for recalibration mechanisms
Arandu/Abaçai (TJAM)	Identifies repetitive and predatory lawsuits through AI-based similarity detection
Ártemis (TJDFT)	Tracks and flags cases of <i>lis pendens</i> , prevention, and <i>res judicata</i> in real time
Bastião (TJPE)	Integrates multiple systems to classify and manage predatory legal demands early
Sherlock System (TJPE)	Real-time detection of repetitive and predatory claims
Blind System (TJPE)	Uses NLP to prevent abuse and detect fraudulent cases
Genesis System (TJPE)	Identifies and isolates repetitive/aggressive litigation

the preliminary filtering of cases, thereby reducing the manual workload and minimising human error.

SRQ4: What challenges and limitations are AI tools facing in addressing predatory litigation within the Brazilian legal context? AI applications face challenges in balancing efficiency with ethical considerations such as fairness and transparency. Studies indicate potential biases in AI algorithms, especially regarding socioeconomic factors affecting litigants' access to justice. Furthermore, limited interoperability among judicial databases and the need for standardised data for machine learning models impede broader implementation. Ethical issues, particularly around data governance and bias mitigation, remain critical for sustainable AI adoption in judicial contexts.

SRQ5: What future opportunities exist for improving AI-driven solutions in the fight against abusive litigation in Brazil's judiciary? Future opportunities include refining data-sharing protocols between courts to support AI systems and improve data interoperability. Expansion of ethical guidelines for AI in judiciary applications can address fairness and transparency concerns. Furthermore, adaptive learning models, which can evolve with new data on abusive practices, offer potential for enhanced detection accuracy and case management efficiency.

Conclusion

The findings of this study highlight the significant role of artificial intelligence in identifying and mitigating repetitive and abusive litigation within the Brazilian judiciary. Various AI-driven initiatives, such as the Genesis, Sherlock, and Blind systems, demonstrate the capacity of machine learning algorithms to detect patterns indicative of predatory legal actions, thus preventing these cases from overburdening the judicial system. The literature and technical reviews indicate that these tools enhance the efficiency of case management and support the integrity and accessibility of justice by filtering frivolous or aggressive demands early in the judicial process.

However, the study also uncovers challenges, including AI's ethical and legal implications, such as potential biases in decision-making and data governance issues. To address these, future work should focus on developing ethical frameworks and refining data-sharing protocols to facilitate AI's responsible and transparent use in legal contexts. Additionally, adaptive AI models that evolve with new data on abusive practices could further improve the accuracy and effectiveness of judicial processes.

In conclusion, integrating AI in the Brazilian judiciary is a promising avenue for enhancing judicial efficiency, provided that ethical and technical challenges are continually addressed. This study contributes to the ongoing discourse on

AI's application in law, underscoring its potential to reform judicial practices and mitigate systemic challenges posed by abusive litigation.

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